

Original Method: Data-Driven Approximation of Earth–Mars Pork-Chop Surfaces

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Abstract—This proposal presents an original method that integrates data-driven modeling with classical interplanetary trajectory analysis. A surrogate model will be trained on synthetically generated Earth–Mars transfer data to approximate optimal ΔV and time-of-flight (TOF) values. The goal is to reproduce the characteristic *pork-chop surface*—the map of launch windows, ΔV , and TOF—while significantly reducing computational effort compared to traditional solvers. This aligns with recent efforts to replace traditional solvers with neural-network surrogates for trajectory optimization and robust design under uncertainty [1]–[3].

Index Terms—pork-chop plot, trajectory optimization, Earth–Mars transfer, surrogate modeling, interplanetary dynamics

I. MOTIVATION

Designing interplanetary trajectories involves solving complex nonlinear equations to determine feasible transfers between planets. Traditional tools such as Lambert solvers require repeated iterations to search launch windows, increasing computational load [1]. The resulting trade space between departure date, arrival date, ΔV , and TOF is visualized using a *pork-chop plot*, which is still a standard tool in preliminary interplanetary mission design.

Recent literature shows success using neural networks and reinforcement learning to approximate or optimize trajectory costs without repeatedly solving differential equations. Ruff *et al.* demonstrate how surrogate neural networks can approximate simulation-based trajectory optimization [2], while Federici *et al.* and Zavoli *et al.* apply meta-reinforcement learning to robust trajectory design under multiple uncertainties [1], [3]. These works motivate the current project to explore an Earth–Mars surrogate model targeting rapid pork-chop surface recreation.

II. EXPECTED CONTRIBUTION

This project will produce a reproducible workflow that:

- Generates a synthetic Earth–Mars transfer dataset using numerical trajectory solutions based on Lambert’s problem [2].
- Trains a surrogate model to approximate ΔV and TOF from input geometry and timing data.
- Documents training and validation using Google Colab tutorials with full reproducibility and clear structure.

- Evaluates model performance using visual and quantitative metrics, aligned with standards in surrogate-based trajectory design research [2], [3].

III. EXPECTED CONTRIBUTION AGAINST COMPETITIVE METHODS

TABLE I
COMPARISON WITH EXISTING METHODS

Method	Key Feature	Limitation	New Element in This Project
Classical Lambert Solver	Analytic, deterministic	Requires iteration per case	Learns entire ΔV surface
Direct Optimization	Accurate baseline	Slow over large launch windows	Provides labeled reference data for training
Surrogate NN [2]	Fast cost predictions	Requires dataset setup	Predicts pork-chop trade space directly
Meta-RL [1], [3]	Handles uncertainties via learned policies	Complex training and deployment	Surrogate trades some accuracy for speed and simplicity

IV. DRAFT FLOWCHART / PSEUDOCODE

Workflow Overview:

- 1) Generate Earth–Mars transfer samples using a Lambert solver.
- 2) Record corresponding ΔV and TOF for each valid trajectory.
- 3) Store data in structured files and split into training, validation, and testing sets.
- 4) Train a surrogate model using appropriate optimization and validation steps.
- 5) Evaluate prediction accuracy and timing efficiency.
- 6) Test on launch opportunities beyond the training window.

To make the workflow clearer and to satisfy the project requirement for a flowchart, Fig. 1 summarizes the proposed pipeline from data generation to surrogate evaluation.

V. CODE IMPLEMENTATION

The project will be implemented in **Google Colab**, following the required tutorial format. Each tutorial will include:

- Documentation using markdown text and headings.
- Installation and version notes for all required packages.

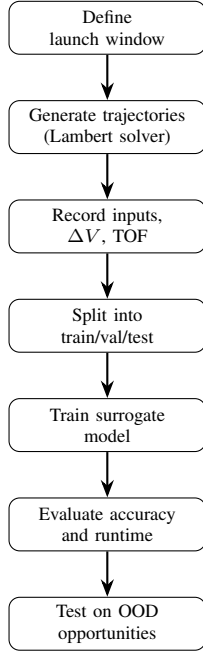


Fig. 1. Flowchart of the proposed Earth-Mars pork-chop surrogate workflow.

- Clear explanations of dataset setup, model training, and testing steps.
- Approximate runtime information and hardware specifications.

This organization mirrors reproducible practices in recent surrogate model development for trajectory planning [2].

VI. DATASET REQUIREMENTS

A. Source and Description

A synthetic dataset will be generated by solving Earth-Mars transfer cases across multiple synodic periods using a two-body approximation, consistent with prior surrogate modeling work [2].

- **Inputs:** departure epoch, arrival epoch, planetary longitudes, and synodic phase angle.
- **Outputs:** total ΔV [km/s], TOF [days].
- **Size:** approximately 1000 samples.
- **Format:** stored as .csv files in training, validation, and testing subdirectories.

B. Splitting and Independence

Data will be split 70% for training, 15% for validation, and 15% for testing. An additional out-of-distribution (OOD) set from a distinct synodic period will test generalization, similar to robustness checks in meta-RL trajectory work [3].

C. Data Documentation Figure

Figure 2 presents a synthetic Earth-Mars **pork-chop plot**, which visualizes contours of total ΔV as a function of departure epoch and time-of-flight. The repeating low- ΔV windows correspond to Hohmann-like transfer opportunities occurring roughly every 26 months. The surrogate model aims

to reproduce these low-cost corridors in a data-driven fashion, inspired by the surrogate mapping of cost landscapes in [2].

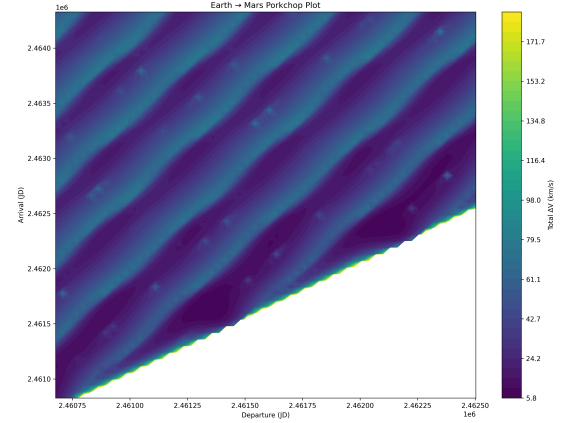


Fig. 2. Synthetic Earth-Mars pork-chop surface showing ΔV contours (color) as a function of departure epoch and TOF. Blue regions indicate low-energy transfer windows.

VII. DATA TRAINING AND INDEPENDENT TESTING METHOD

The surrogate model will be trained on the primary dataset and independently tested on unseen launch configurations. Performance will be evaluated using quantitative error metrics and qualitative visual comparisons to the baseline pork-chop surface. This structure is analogous to the workflows in Ruff *et al.* [2], where surrogates are validated on trajectory-planning tasks not seen during training.

VIII. VALIDATION METRICS

- Mean Absolute Error (MAE) [km/s]
- Root Mean Squared Error (RMSE)
- Coefficient of Determination (R^2)
- Relative evaluation time compared to the numerical solver

These metrics are standard in surrogate-based optimization studies and help compare accuracy and efficiency trade-offs [2].

IX. EXPECTED FLOW AND CONNECTIONS

The project demonstrates a clear and structured flow:

- 1) Synthetic pork-chop dataset generation provides labeled examples.
- 2) The surrogate model approximates the relationship between launch parameters and ΔV , supporting fast trade-space exploration.
- 3) Validation metrics measure both predictive accuracy and computational efficiency, echoing evaluation strategies from surrogate and RL-based trajectory studies [1]–[3].
- 4) Results and methodology are documented through six Google Colab tutorials covering dataset setup, model description, optimization, testing, refinement, and full workflow execution.

X. CONCLUSION

This proposal outlines a data-driven approach to approximating Earth–Mars pork-chop surfaces using a surrogate model trained on numerically generated transfer data. By drawing inspiration from reinforcement learning approaches to robust trajectory design [1], [3] and from surrogate neural networks for simulation-based optimization [2], the project aims to demonstrate how even a relatively small-scale surrogate can provide meaningful runtime reductions and intuitive visualizations for early mission design.

REFERENCES

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